

Advanced VLSI Design - Lab 3

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1 Layout and Schematic of a 4-Pin NMOS Transistor

To make a simple 4-pin NMOS transistor in Electric VLSI, the 4-pin NMOS was selected in the schematic components, and the width was set to 10 and length set to 2. The source, gate, drain, and body (bulk) were labelled using *Off-Page* nodes exported with their corresponding letters. It is important to note that in a NMOS transistor, the body (bulk) is tied to the lowest potential (ground), whereas in a PMOS transistor, the body (bulk) is tied to the highest potential (VDD). A spice model was set for the NMOS transistor and labeled according to the file *C5_models.txt*, which sets simulation parameters for the NMOS that was designed. The schematic design of the NMOS with corresponding IC can be seen in *figure 1*.

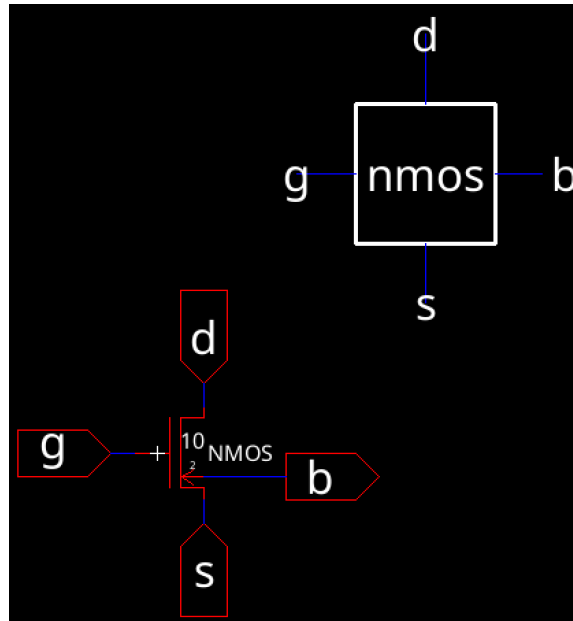


Figure 1: Schematic View of NMOS Design with Corresponding Icon

Building the NMOS layout took a few steps to ensure successful completion. An NMOS node was placed at the center points (0,0) of the layout and set with a width of 10 and length of 2

to match the transistor designed in the schematic. Two nActive nodes were placed on either end (top and bottom) of the NMOS transistor and set to the same dimension. The aim for the initial transistor design was to get the nActive nodes as close to the NMOS as possible; however, it was important that these nodes were connected beforehand to ensure no errors or floating points. Once connected between the nActives and NMOS transistor were made, they were grouped as closely as possible before the file log read any errors.

As stated in the schematic paragraph, it is important that the body (bulk) is not forgotten on the layout. To do this, a pWell node was added, separated slightly from the transistor, and given the same dimensions as the nActives and NMOS transistor. A metal-1 to polysilicon-1 node was placed for the gate, and five total metal-1 pins corresponding to ground, drain, source, body, and ground. The layout for the NMOS can be seen in figure 2.

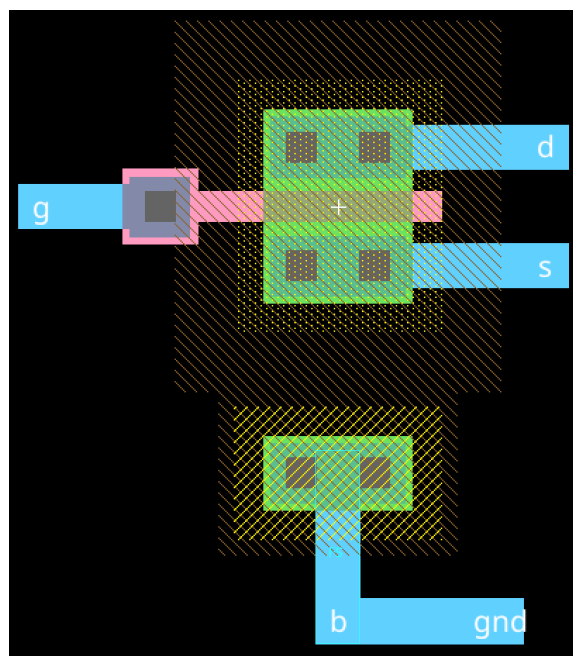


Figure 2: Layout View of NMOS Design

2 Layout and Design of Diode Connections for ESD Protection

In a simple ESD (electro-static discharge) circuit, two diodes are used to protect the circuitry from high voltage spikes coming from the pad. In this configuration, the anode of diode 1 is connected to the inout pin and the pActive, while the cathode is connected to VDD and the nWell. The anode of diode 2 is connected to ground and the pWell, while the cathode is connected to the inout pin and nActive. This section demonstrates the schematic and layout of individual diodes, and the

corresponding ESD protection circuitry connecting to a pad.

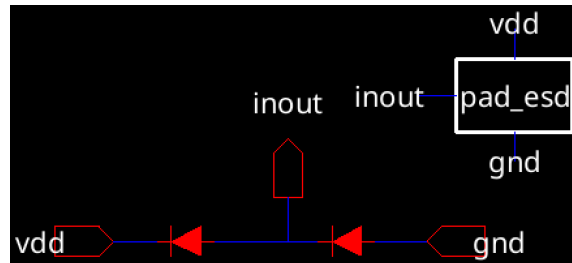


Figure 3: Schematic View of Pad with ESD Protection Diodes and Corresponding Icon

The nActive - pWell diode began by using a simple diode symbol in the schematic components, attaching *Off-Page* nodes, and labeling the pins accordingly with pWell at the anode and nActive at the cathode.

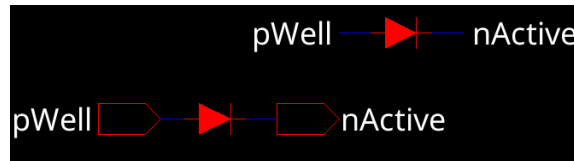


Figure 4: Schematic View of nActive - pWell with Corresponding Icon

Following this, the pActive - nWell diode was made using the same method, but labelling the anode as pActive and cathode as nWell.

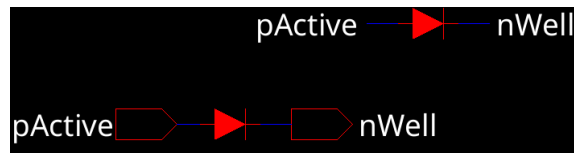


Figure 5: Schematic View of pActive - nWell with Corresponding Icon

To build the layout for both diodes, all that was needed is a simple PN-junction. For the nActive - pWell, a metal-1 to nActive node was placed and sized at 50 x 50, and the same was done for the metal-1 to pWell node. The two nodes were then placed as closely together as possible and metal connects and export names were added. The same procedure was done to make the pActive - nWell.

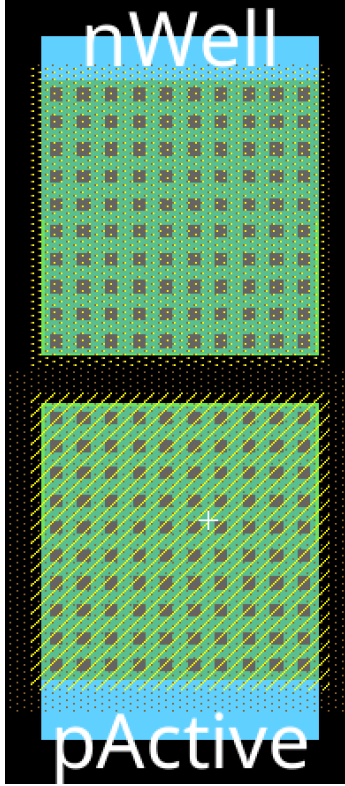


Figure 6: Layout View of pActive - nWell PN-Junction

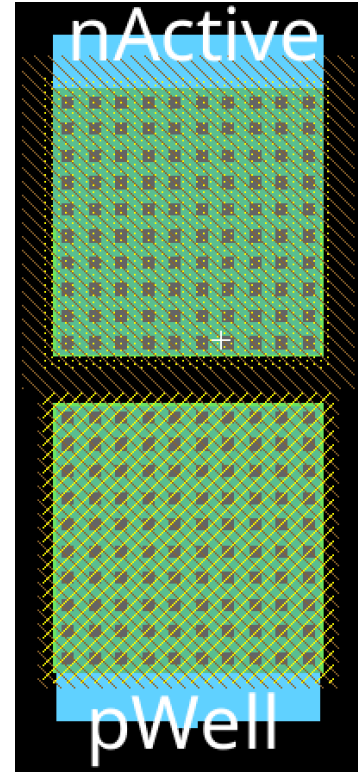


Figure 7: Layout View of nActive - pWell PN-Junction

To build the layout for the ESD protection pad, the pWell - nActive diode was placed on the left side of the pad, while the nWell - pActive diode was placed on the right. A metal 2 pin was placed in between the diodes and set as the **inout** pin for connection between the NMOS and pad. Four metal-1 pins were used to create a **vdd** and **gnd** rail used to connect the 8-pins together. Since the pWell is connected to ground, two metal-1 to metal-2 connections were placed to allow the ground line to run over the high potential line.

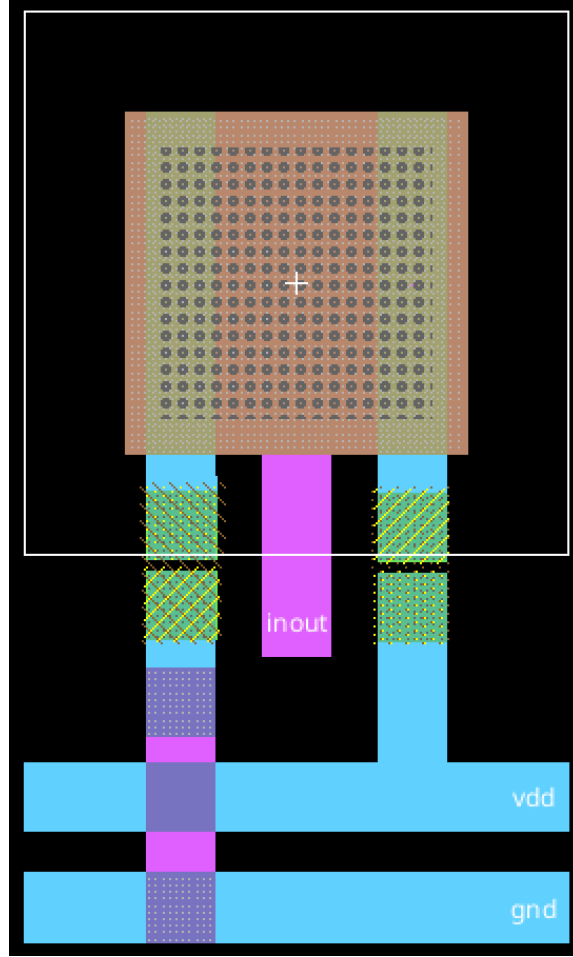


Figure 8: Layout View of Singular Pad with ESD Protection

3 Layout and Schematic of an ESD Pad-Frame

To create the schematic for the ESD pad-frame, the icon of the single pad was placed in the new schematic view and nodes were connected accordingly. The **inout** pin was labelled **pin[1:8]** and linked via a bus connection to signify multiple lines.

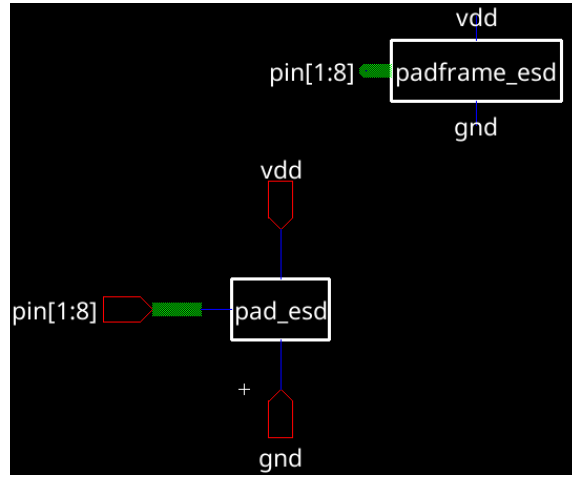


Figure 9: Schematic View of Pad-Frame with ESD Protection

Compared to the simplified pad-frame design completed in Lab 2, the array function cannot be used effectively to create the new pad-frame, as the ESD circuitry had to be lined up correctly. To compensate for this, a 2x2 array was made which allowed the left-most-half pins to be perfectly aligned. These pins were then copied and pasted onto the layout for a total of 8-pins. Effort was put into aligning all the pins such that there was even geometry all around. Once this was achieved, the pins were wired to their respective connection points labelled with their respective export names.

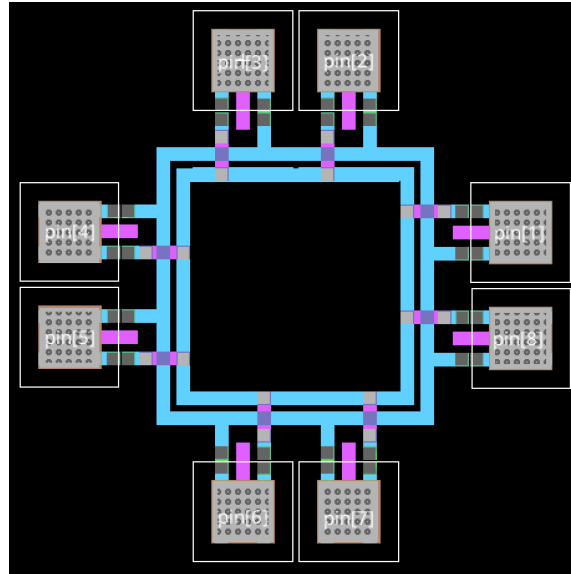


Figure 10: Layout View of 8-Pin Padframe with ESD Protection

4 Layout and Schematic of the Final IC with ESD Protection

To build the schematic of the final IC with ESD protection, the pad-frame ESD icon and NMOS icon were placed into the new schematic. Wires were drawn from each pin; however, no exports were used and the lines were simply renamed according to their designated pin connections.

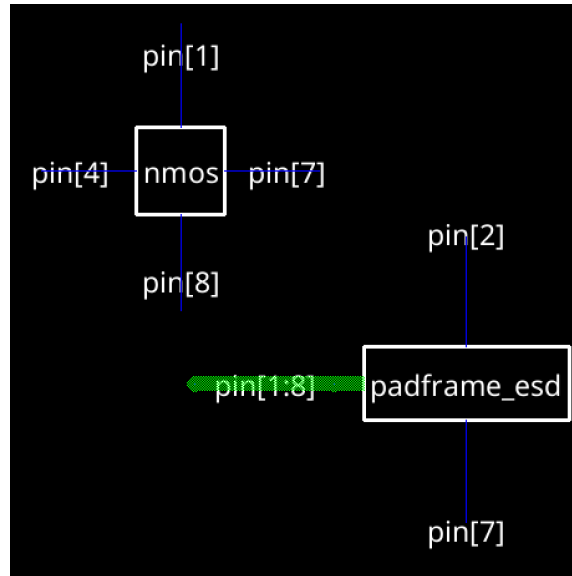


Figure 11: Schematic View of Final IC with ESD Protection

To build the layout of the final IC with ESD protection, the ESD pad-frame was first placed into the new layout view, followed by the NMOS transistor placed in the center of the pad-frame. To make connections between the transistor and pad-frame easier, the pad-frame was set to hard-to-select. Metal-1 to metal-2 connections were placed in front of their respective pins corresponding to the pinouts on the transistor. Once done, all connections were made using 3-5 width metal-1 wire and 20 width metal-2 wire.

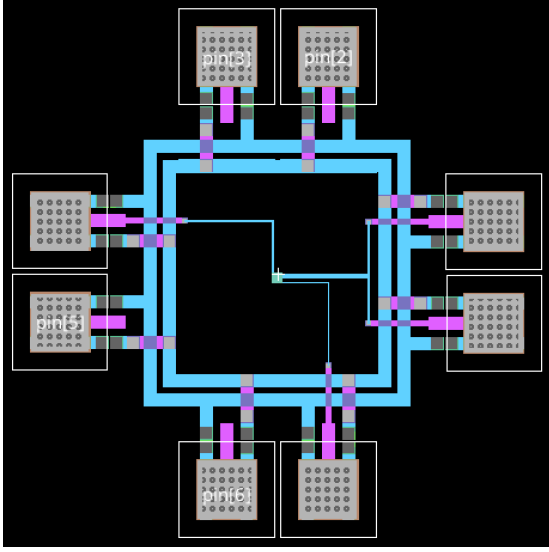


Figure 12: Layout View of Final IC with ESD Protection

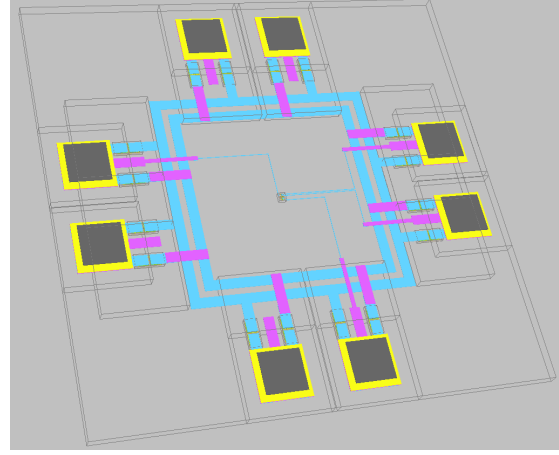


Figure 13: 3D Render of Final IC with ESD Protection

5 Comparative Design of No ESD Protection vs. ESD Protection

Lab 2 utilized a pad-frame with no ESD protection, and a design of the IC with no ESD protection was presented in this lab to demonstrate the comparative differences in design. Both systems use the identical NMOS discusses in section 1.

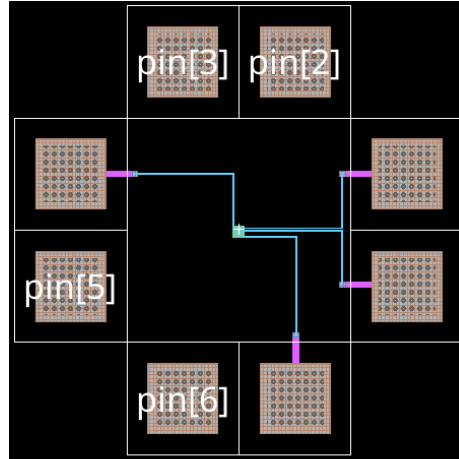


Figure 14: Layout View of Final IC with No ESD for Comparative Reference.

6 Documentation

[1] - ENCE_3501_Lab_3.pdf

[2] - Lab_1_NMOS_ESD.pdf

[3] - lab_3_nmos_esd_template.jelib